

Certificate no: Oxygen Optode 4831\_216\_251683  
FoillID: 2310M

Product: Oxygen Optode 4831  
Serial no: 216  
Calibration date: 17.03.2026

## 2-POINT RECALIBRATION, MULTIPOINT CALIBRATED OXYGEN OPTODE

This is to certify that this product has been calibrated and verified using the following reference equipment:

| Reference Equipment                    | Serial                                  |
|----------------------------------------|-----------------------------------------|
| Calibration Bath Model Toho TM-005-P-A | 12-K                                    |
| Fluke 5615 PRT                         | Serial No. 802054 and Serial No. 849155 |
| Fluke CHUB E-4                         | Serial No. A7C677                       |
| Reference sensor 4330                  | 338                                     |

### Specification

|                     | O2 Concentration            | Air Saturation    | Temperature           |
|---------------------|-----------------------------|-------------------|-----------------------|
| Range               | 0-500 µM                    | 0 - 120%          | -5 to +40°C           |
| Accuracy            | < ±3µM or ±2% <sup>1)</sup> | ±2% <sup>2)</sup> | ±0.03°C <sup>3)</sup> |
| Resolution          | < 1 µM                      | < 0.4%            | 0.01°C                |
| Settling Time (63%) | < 25 seconds                | < 25 seconds      | < 2 seconds           |

1) Requires salinity compensation for salinity variations > 1mS/cm, and pressure compensation for pressure > 100meter

2) Within calibrated range 0 - 120% / 0 - 30°C

3) Within calibrated range 0 - 36°C for standard version and 0 - 30°C for multipoint version

### Calibration points and readings:

|                          | Air Saturated Water | Zero Solution (Na2SO3) |
|--------------------------|---------------------|------------------------|
| Phase reading (°)        | 33.75               | 60.66                  |
| Temperature reading (°C) | 9.80                | 13.39                  |
| Air Pressure (hPa)       | 1004.69             |                        |

### Giving these coefficients

| Index    | 0 (Offset)    | 1 (Slope)    |
|----------|---------------|--------------|
| ConcCoef | -5.463217E+00 | 1.107534E+00 |

### With following settings

| Index            | 0            | 1             | 2            | 3             | 4            | 5            | 6           |
|------------------|--------------|---------------|--------------|---------------|--------------|--------------|-------------|
| SVUFoilCoef      | 2.75399E-03  | 1.17456E-04   | 2.27721E-06  | 1.66602E+02   | -1.98008E-01 | -4.14581E+01 | 3.32436E+00 |
| PhaseCoef        | 0.000000E+00 | 1.000000E+00  | 0.000000E+00 | 0.000000E+00  |              |              |             |
| TempCoef         | 2.446417E+01 | -3.147211E-02 | 3.017436E-06 | -4.299026E-09 | 0.000000E+00 | 0.000000E+00 |             |
| Salinity (mS/cm) | 0.00         |               |              |               |              |              |             |
| Firmware Version | 5.3.1        |               |              |               |              |              |             |

Date:17.03.2026

  
Sr. Repair Technician